

Amirhossein Nafissi

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EDUCATION

University of Waterloo | BSE, Software Engineering

Sep. 2025 – Jan. 2030

University of Guelph | B Computing, Software Engineering (Transferred); GPA: 4.0/4.0

Sep. 2024 – Apr. 2025

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, TypeScript, HTML, CSS, ARM64 Assembly, RISC-V Assembly, Verilog

Frameworks & Libraries: Pandas, NumPy, Matplotlib, React, Node.js, Express.js

Tools, Hardware & Databases: Git, Linux, PostgreSQL, MongoDB, Claude Code, Pi Agent, FPGA, Raspberry Pi

EXPERIENCE

TechOps Developer

May 2026 – August 2026

Instant Risk Coverage (IRC)

Toronto, ON

- Built a **Claude Code** powered **PR-review** visualization layer generating import-impact, module-dependency, and function-call graphs with cycle detection, including built-in self-tests. Reducing the time it takes to review PRs.
- Audited the internal **agentic coding** knowledge index, reproducing 22 defects, each with line citations and a runnable verification command; delivered remediation findings to senior engineering to reduce AI hallucinations, cut token usage, sharpen **LLM** output, and improve the **developer experience**.
- Developed xrefscope, a dependency-free **SQLite**-to-HTML viewer rendering the agentic coding knowledge index as a static offline page; improving database visibility and accelerating **development workflow**.
- Engineered the platform's first USD payment support, routing Stripe account, currency, and tax through a **MongoDB** flag and a generated **deploy matrix (Python, GitHub Actions, Docker)**. **Shipped to production** and in active use, increasing the platform's user reach to US customers.
- Developed the **production Mage/PostgreSQL ETL pipeline** for e-commerce order **reporting**, migrating order data from a legacy platform; adding idempotent upserts and fan-out guards.
- Fast-tracked the creation of the organization-wide **AI governance handbook** by authoring key structural sections and leading consensus-building, resolving 73 of 77 inline review threads (95%) to secure stakeholder approval.
- Supported TechOps operations for insurance portals, including client onboarding, cancellations, debugging, and iterative reliability improvements.

Founding Member & Software Developer

Nov. 2025 – Present

WatQ – Waterloo Quantum Design Team

Waterloo, ON

- Designed and built the team's public-facing website using **Next.js**, **React** Three Fiber, and Framer Motion featuring interactive elements, increasing the design team's outreach.
- Architected the team's Notion workspace as the single source of truth, reducing information fragmentation and accelerating onboarding.

Machine Learning Intern

Jul. 2023 – Oct. 2023

University of Waterloo

Waterloo, ON

- Worked with a research group to collect a sensitive security dataset for training a robust **machine learning** model.
- Gained training and hands-on experience related to creating simple machine-learning models with various **Python** libraries to be used in personal projects.

PROJECTS

CueMate | Kotlin, Jetpack Compose, MediaPipe, CameraX

- Developed a privacy-first Android assistive application that leverages **on-device computer vision** to interpret facial expressions and hand gestures for visually impaired users.
- Integrated **Google's MediaPipe** framework to run real-time facial landmark analysis and gesture recognition locally, providing multimodal feedback via Text-to-Speech and directional haptics.

Visualize Quantum | Next.js, TypeScript, Python, NumPy/SciPy, Qiskit

- Engineered a modular quantum algorithm playground with a plugin architecture and central registry for seamless algorithm integration, aiding in quantum education.
- Implemented a Graph Coloring solver using custom NumPy QAOA (COBYLA, INTERP) with local simulation and IBM Quantum hardware execution, plus interactive visualizations deployed via Vercel serverless Python functions.

HONORS & AWARDS

Schulich Leader Scholarship: High School Nominee

Goose Hacks: Ranked 4th best project out of 190+ participants